

OE-EC704B — Optimization Technique

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Module 5 Exam Notes: Constrained Optimization Algorithms

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1. Characteristics of Constrained Problems

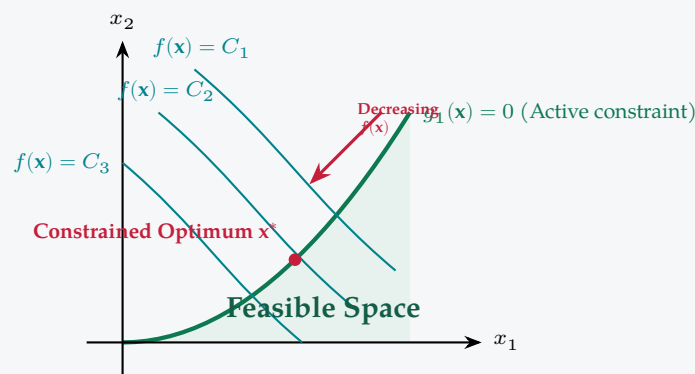
Constrained optimization restricts the search space to a feasible region. Let us analyze the constraint boundaries:

Active vs. Inactive Constraints

At a candidate design point $\mathbf{x}$:

- An inequality constraint $g_j(\mathbf{x}) \leq 0$ is said to be **Active** if $g_j(\mathbf{x}) = 0$. The point lies exactly on the boundary of that constraint.
 - It is **Inactive** if $g_j(\mathbf{x}) < 0$. The constraint is satisfied, and the point lies in the interior of the feasible space.
 - It is **Violated** if $g_j(\mathbf{x}) > 0$. The design point is infeasible.
- All equality constraints $h_k(\mathbf{x}) = 0$ must be active at all feasible points.

Feasible Region and Constrained Optimum



2. Direct Constrained Optimization Methods

Direct methods evaluate only function values and directly handle constraints by adjusting candidate coordinates.

2.1 Box's Complex Method

Box's Complex Method is an extension of the Nelder-Mead Simplex method designed to handle inequality constraints:

- Uses a geometric shape called a **Complex** consisting of $k \geq n + 1$ vertices (typically $k = 2n$).
- If a generated reflection point violates an inequality constraint, it is moved halfway toward the centroid of the remaining feasible points until it becomes feasible.

2.2 Kelly's Cutting Plane Method

Kelly's Cutting Plane Method solves non-linear convex constrained problems:

1. Linearizes all non-linear constraints using first-order Taylor expansions to form a linear programming (LP) relaxation.
2. Solves the LP relaxation. If the solution is infeasible for the original non-linear constraints, it adds a new linear inequality constraint (a **Cutting Plane**) that slices off the infeasible LP solution.
3. Repeats the process until convergence, generating a sequence of tightening linear hulls.

3. Indirect Methods (Transformation Techniques)

Transformation techniques reformulate a constrained problem into an equivalent sequence of unconstrained problems.

3.1 Exterior Penalty Function Method

The exterior penalty method starts search in the infeasible region and penalizes constraint violations, forcing iterations toward the feasible boundary.

Exterior Penalty Function Formulation

The unconstrained penalty objective function Φ is defined as:

$$\Phi(\mathbf{x}, r_p) = f(\mathbf{x}) + r_p \sum_{j=1}^m [\max(0, g_j(\mathbf{x}))]^2 + \frac{1}{r_p} \sum_{k=1}^p [h_k(\mathbf{x})]^2 \quad (1)$$

Where r_p is a positive penalty parameter.

Algorithm:

1. Select an initial penalty value $r_p^{(1)}$ (usually small) and minimize $\Phi(\mathbf{x}, r_p^{(1)})$ unconstrained.
2. Increase the parameter: $r_p^{(k+1)} = c \cdot r_p^{(k)}$ (where $c > 1$).
3. Re-minimize Φ . As $r_p \rightarrow \infty$, the sequence of unconstrained minima converges to the constrained optimum $\mathbf{x}^*$.

3.2 Interior Penalty (Barrier) Function Method

The interior penalty method starts inside the feasible region and creates a barrier at the boundary that prevents iterations from escaping into the infeasible space.

Interior Barrier Function Formulations

The unconstrained barrier objective function Φ is defined using one of two barriers:

- **Inverse Barrier Function:**

$$\Phi(\mathbf{x}, r_k) = f(\mathbf{x}) - r_k \sum_{j=1}^m \frac{1}{g_j(\mathbf{x})} \quad (2)$$

- **Logarithmic Barrier Function:**

$$\Phi(\mathbf{x}, r_k) = f(\mathbf{x}) - r_k \sum_{j=1}^m \ln [-g_j(\mathbf{x})] \quad (3)$$

Where r_k is the positive barrier parameter.

Algorithm:

1. Start at a strictly feasible point. Minimize $\Phi(\mathbf{x}, r_k^{(1)})$.
2. Decrease the parameter: $r_k^{(k+1)} = d \cdot r_k^{(k)}$ (where $0 < d < 1$).
3. Re-minimize. As $r_k \rightarrow 0$, the barrier effect weakens at the boundary, allowing convergence to the boundary optimum $\mathbf{x}^*$.

4. Convex Optimization Methods

Convex optimization is a special class of optimization with unique mathematical guarantees.

Convex Programming Problem

A constrained optimization problem is a **Convex Programming Problem** if:

1. The objective function $f(\mathbf{x})$ is convex.
2. All inequality constraints $g_j(\mathbf{x}) \leq 0$ are convex functions.
3. All equality constraints $h_k(\mathbf{x}) = 0$ are linear (affine) functions.

Theorem: For a convex programming problem, **any local minimum is guaranteed to be the global minimum**, eliminating the risk of getting trapped in local sub-optimal states.

Quick Revision — Module 5

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Key Concepts & Formulations

Active Inequality	$g_j(\mathbf{x}^*) = 0$ (optimum lies on the constraint boundary).
Inactive Inequality	$g_j(\mathbf{x}^*) < 0$ (constraint is satisfied with slack).
Exterior Penalty Φ	$\Phi = f(\mathbf{x}) + r_p \sum [\max(0, g_j)]^2$ (where $r_p \rightarrow \infty$).
Inverse Interior Barrier	$\Phi = f(\mathbf{x}) - r_k \sum \frac{1}{g_j}$ (where $r_k \rightarrow 0$).
Logarithmic Barrier	$\Phi = f(\mathbf{x}) - r_k \sum \ln(-g_j)$ (where $r_k \rightarrow 0$).
Convex Definition	$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})$ for $\lambda \in [0, 1]$.

One-Line Definitions

- **Active Constraint:** A constraint whose boundary passes directly through the current design point.
- **Complex Method:** A constrained direct search method utilizing $2n$ vertices to explore bounded spaces.
- **Transformation Technique:** Reformulating constrained problems into sequential unconstrained formulations.
- **Exterior Penalty Method:** A transformation technique that approaches the boundary from the infeasible design region.
- **Interior Barrier Method:** A transformation technique that approaches the boundary from strictly within the feasible region.
- **Convex Function:** A function where a line segment connecting any two points on its graph lies above or on the graph.

Frequently Asked / Viva Questions

Q1. What is the difference between active and inactive constraints at an optimum point?

An active constraint satisfies $g_j(\mathbf{x}^*) = 0$ (lies on the boundary). An inactive constraint satisfies $g_j(\mathbf{x}^*) < 0$ (lies inside the boundary) and does not limit the local optimum.

Q2. Compare the search paths of the Exterior Penalty and Interior Barrier methods.

The exterior penalty method starts from an infeasible point and moves inward toward the feasible boundary as $r_p \rightarrow \infty$. The interior barrier method starts from a strictly feasible point and moves outward toward the boundary as $r_k \rightarrow 0$.

Frequently Asked / Viva Questions (contd.)

Q3. Why does the Interior Barrier method require a strictly feasible starting point?

Because the barrier terms (such as $1/g_j$ or $\ln(-g_j)$) are mathematically undefined or return complex/imaginary values in the infeasible region ($g_j(\mathbf{x}) > 0$).

Q4. State Sylvester's conditions for a function to be strictly convex.

A function is strictly convex if its Hessian matrix $H(\mathbf{x})$ is positive definite (all eigenvalues > 0) at all points in the design space.

Q5. What is the main advantage of Convex Programming?

Any local minimum of a convex programming problem is guaranteed to be the global minimum.

Q6. Explain the concept behind Kelly's Cutting Plane method.

It linearizes the non-linear constraints to solve an approximating linear program. It then iteratively adds linear constraint inequalities (cutting planes) to cut off infeasible areas.

Q7. Why are equality constraints challenging for the Interior Barrier method?

Because equality constraints $h_k(\mathbf{x}) = 0$ define a region with zero thickness, leaving no "interior" space to start a strictly feasible search path.

Q8. How does Box's Complex method handle constraint violations?

If a generated point is infeasible, it is iteratively retracted halfway toward the centroid of the remaining feasible points until it becomes feasible.

Q9. Write down the logarithmic barrier objective function expression.

$$\Phi(\mathbf{x}, r_k) = f(\mathbf{x}) - r_k \sum_{j=1}^m \ln(-g_j(\mathbf{x})).$$

Q10. What is the physical meaning of the penalty parameter r_p in the exterior penalty method?

It acts as a scaling coefficient for constraint violations. As $r_p \rightarrow \infty$, the penalty for violating constraints becomes extremely large, forcing the unconstrained optimizer to find a feasible solution.